

Step 1: Assign ip address for all Routers and static IP's

```

R1
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    1.0.0.0/8 is directly connected, Ethernet0/1
L    1.0.0.1/32 is directly connected, Ethernet0/1
S    2.0.0.0/8 [1/0] via 1.0.0.2
S    3.0.0.0/8 [1/0] via 1.0.0.2
S    4.0.0.0/8 [1/0] via 1.0.0.2
S    5.0.0.0/8 [1/0] via 1.0.0.2
S    6.0.0.0/8 [1/0] via 1.0.0.2
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
S    135.82.0.0/16 [1/0] via 1.0.0.2
S    153.69.0.0/16 [1/0] via 1.0.0.2
S    182.46.0.0/16 [1/0] via 1.0.0.2
S    200.10.156.0/24 [1/0] via 1.0.0.2
R1#
R1#
R1#

```

```

R2
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    1.0.0.0/8 is directly connected, Ethernet0/2
L    1.0.0.2/32 is directly connected, Ethernet0/2
2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    2.0.0.0/8 is directly connected, Ethernet0/1
L    2.0.0.1/32 is directly connected, Ethernet0/1
S    3.0.0.0/8 [1/0] via 2.0.0.2
S    4.0.0.0/8 [1/0] via 2.0.0.2
S    5.0.0.0/8 [1/0] via 2.0.0.2
6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    6.0.0.0/8 is directly connected, Ethernet0/3
L    6.0.0.1/32 is directly connected, Ethernet0/3
S    10.0.0.0/8 [1/0] via 1.0.0.1
S    135.82.0.0/16 [1/0] via 2.0.0.2
S    153.69.0.0/16 [1/0] via 6.0.0.2
S    182.46.0.0/16 [1/0] via 2.0.0.2
200.10.156.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.10.156.0/24 is directly connected, Ethernet0/0
L    200.10.156.1/32 is directly connected, Ethernet0/0
R2#

```

R3

a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 2.0.0.1
C    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    2.0.0.0/8 is directly connected, Ethernet0/0
L    2.0.0.2/32 is directly connected, Ethernet0/0
C    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    3.0.0.0/8 is directly connected, Ethernet0/1
L    3.0.0.1/32 is directly connected, Ethernet0/1
C    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/2
L    4.0.0.1/32 is directly connected, Ethernet0/2
S    5.0.0.0/8 [1/0] via 4.0.0.2
S    6.0.0.0/8 [1/0] via 2.0.0.1
S    10.0.0.0/8 [1/0] via 2.0.0.1
S    135.82.0.0/16 [1/0] via 4.0.0.2
S    153.69.0.0/16 [1/0] via 2.0.0.1
S    182.46.0.0/16 [1/0] via 3.0.0.2
S    200.10.156.0/24 [1/0] via 2.0.0.1
```

R3#

R3#

R4

a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 3.0.0.1
S    2.0.0.0/8 [1/0] via 3.0.0.1
C    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    3.0.0.0/8 is directly connected, Ethernet0/1
L    3.0.0.2/32 is directly connected, Ethernet0/1
S    4.0.0.0/8 [1/0] via 3.0.0.1
S    5.0.0.0/8 [1/0] via 3.0.0.1
S    6.0.0.0/8 [1/0] via 3.0.0.1
S    10.0.0.0/8 [1/0] via 3.0.0.1
S    135.82.0.0/16 [1/0] via 3.0.0.1
S    153.69.0.0/16 [1/0] via 3.0.0.1
C    182.46.0.0/16 is variably subnetted, 2 subnets, 2 masks
L    182.46.0.0/16 is directly connected, Ethernet0/0
L    182.46.0.1/32 is directly connected, Ethernet0/0
S    200.10.156.0/24 [1/0] via 3.0.0.1
```

R4#

R4#

R4#

R4#

R5

ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 4.0.0.1
S    2.0.0.0/8 [1/0] via 4.0.0.1
S    3.0.0.0/8 [1/0] via 4.0.0.1
    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/1
L    4.0.0.2/32 is directly connected, Ethernet0/1
    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    5.0.0.0/8 is directly connected, Ethernet0/0
L    5.0.0.1/32 is directly connected, Ethernet0/0
S    6.0.0.0/8 [1/0] via 4.0.0.1
S   10.0.0.0/8 [1/0] via 4.0.0.1
S   135.82.0.0/16 [1/0] via 5.0.0.2
S   153.69.0.0/16 [1/0] via 4.0.0.1
S   182.46.0.0/16 [1/0] via 4.0.0.1
S   200.10.156.0/24 [1/0] via 4.0.0.1
```

R5#

R5#

R6

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 5.0.0.1 to network 0.0.0.0

```
S*   0.0.0.0/0 [1/0] via 5.0.0.1
    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    5.0.0.0/8 is directly connected, Ethernet0/1
L    5.0.0.2/32 is directly connected, Ethernet0/1
    135.82.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    135.82.0.0/16 is directly connected, Ethernet0/0
L    135.82.0.1/32 is directly connected, Ethernet0/0
```

R6#

R6#

R6#

R6#

R6#

R6#

R6#

```
R7
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 6.0.0.1
S    2.0.0.0/8 [1/0] via 6.0.0.1
S    3.0.0.0/8 [1/0] via 6.0.0.1
S    4.0.0.0/8 [1/0] via 6.0.0.1
S    5.0.0.0/8 [1/0] via 6.0.0.1
    6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      6.0.0.0/8 is directly connected, Ethernet0/1
L      6.0.0.2/32 is directly connected, Ethernet0/1
S    10.0.0.0/8 [1/0] via 6.0.0.1
S   135.82.0.0/16 [1/0] via 6.0.0.1
   153.69.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     153.69.0.0/16 is directly connected, Ethernet0/0
L     153.69.0.1/32 is directly connected, Ethernet0/0
S   182.46.0.0/16 [1/0] via 6.0.0.1
S   200.10.156.0/24 [1/0] via 6.0.0.1
R7#
R7#
R7#
```

Step 2: Assigning ip for all Linux servers

```
QEMU (Linux1)
ENABLE=yes
NAME=Ethernet
IPADDRESS=10.0.0.10
CIDRLEN=8
NETWORK=10.0.0.0
NETMASK=255.0.0.0
GATEWAY=10.0.0.1
BROADCAST=10.255.255.255
DHCP=no

- /etc/sysconfig/network-scripts/ifcfg-eth0 1/9 11%
```

```
QEMU (Linux2)
ENABLE=yes
NAME=Ethernet
IPADDRESS=10.0.0.20
CIDRLEN=8
NETWORK=10.0.0.0
NETMASK=255.0.0.0
GATEWAY=10.0.0.1
BROADCAST=10.255.255.255
DHCP=no

...

/etc/sysconfig/network-scripts/ifcfg-eth0 1/9 11%
```

```
QEMU (Linux3)
ENABLE=yes
NAME=Ethernet
IPADDRESS=10.0.0.30
CIDRLEN=8
NETWORK=10.0.0.0
NETMASK=255.0.0.0
GATEWAY=10.0.0.1
BROADCAST=10.255.255.255
DHCP=no

...

/etc/sysconfig/network-scripts/ifcfg-eth0 1/9 11%
```

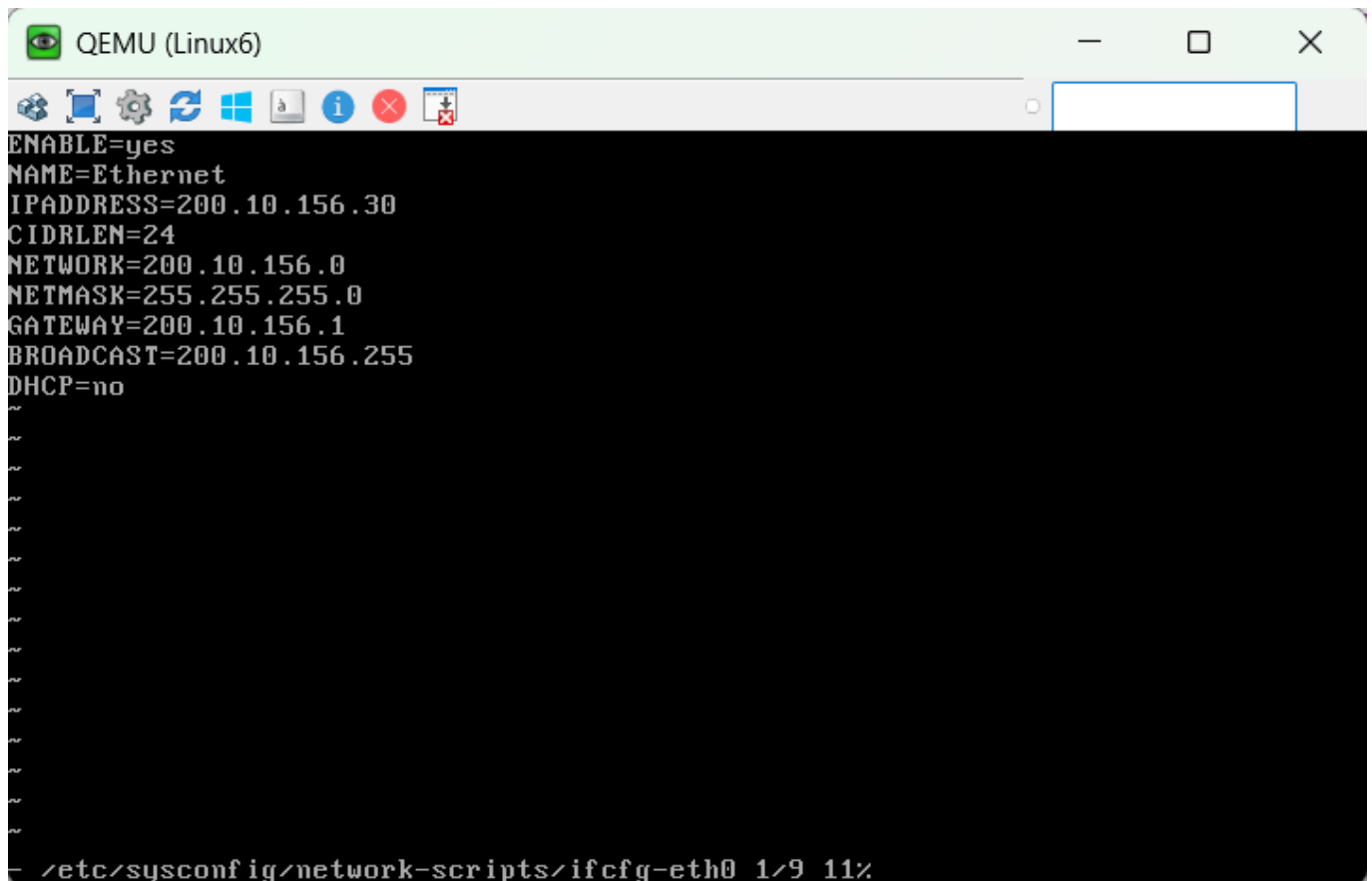
```
QEMU (Linux4)
ENABLE=yes
NAME=Ethernet
IPADDRESS=200.10.156.10
CIDRLEN=24
NETWORK=200.10.156.0

NETMASK=255.255.255.0
GATEWAY=200.10.156.1
BROADCAST=200.10.156.255
DHCP=no

- /etc/susconfig/network-scripts/ifcfg-eth0 1/10 10%
```

```
QEMU (Linux5)
ENABLE=yes
NAME=Ethernet
IPADDRESS=200.10.156.20
CIDRLEN=24
NETWORK=200.10.156.0
NETMASK=255.255.255.0
GATEWAY=200.10.156.1
BROADCAST=200.10.156.255
DHCP=no

- /etc/susconfig/network-scripts/ifcfg-eth0 1/9 11%
```



A screenshot of a QEMU (Linux6) terminal window. The window has a title bar with the QEMU logo and the text "QEMU (Linux6)". Below the title bar is a toolbar with various icons. The terminal itself has a black background with white text. It shows the following configuration:

```
ENABLE=yes
NAME=Ethernet
IPADDRESS=200.10.156.30
CIDRLEN=24
NETWORK=200.10.156.0
NETMASK=255.255.255.0
GATEWAY=200.10.156.1
BROADCAST=200.10.156.255
DHCP=no
```

Below this configuration, there are several lines of tilde (~) characters. At the bottom of the terminal, the command `/etc/sysconfig/network-scripts/ifcfg-eth0 1/9 11%` is visible.

Step 3: Get webpages from pc using ip address





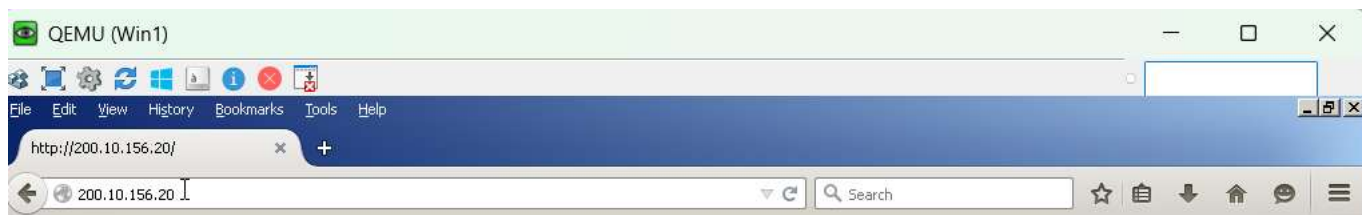
Hello World! This file is located in /home/hthttpd



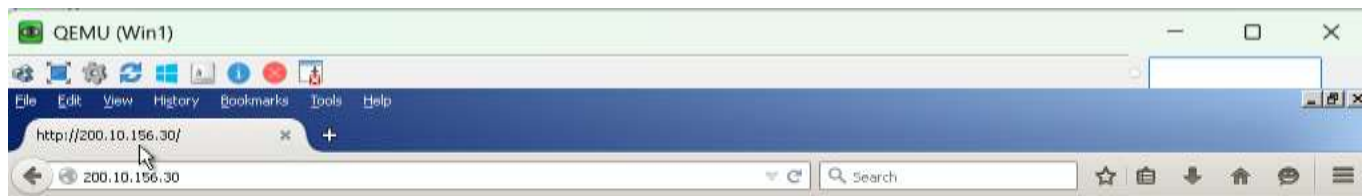
Hello World! This file is located in /home/hthttpd



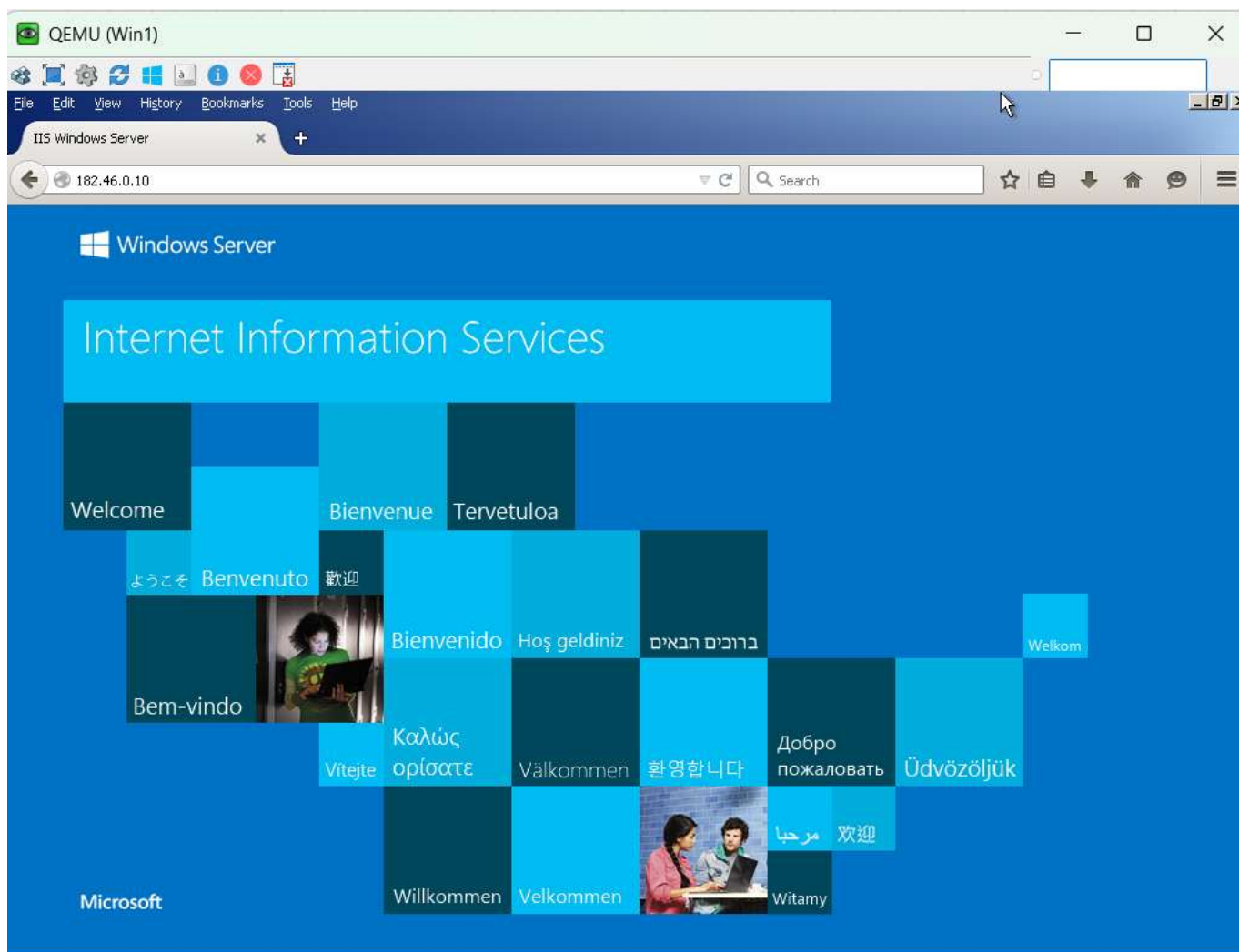
Hello World! This file is located in /home/tthtpd



Hello World! This file is located in /home/tthtpd



Hello World! This file is located in /home/httpd



Step 4 :search web page using Dns

